Chap 6 : Chemical Equilibrium 考点

1. at equilibrium : $rate_{forward} = rate_{reverse}$

2. $K < \begin{matrix} K_c \\ K_p \end{matrix}$ 是用各个物质的分压来算！看清题目给好了分压，还是需用 $X_{mol} \times P_{total}$

$aA + bB \rightleftharpoons cC + dD \qquad K_c = \dfrac{[C]^c[D]^d}{[A]^a[B]^b}$

· 反应相反K取倒数

· 反应相加K相乘

在平衡常数表达式中，只有参与反应的气体和溶液中有一定浓度的物质才会计入K的计算.

solids and liquids are not included in K , but $H_2O(g)$ should be written into K_c & K_p.

3. IUPAC (以后运算都以此形式)

$K_c = \dfrac{([C]/[C]^\circ)^c([D]/[D]^\circ)^d}{([A]/[A]^\circ)^a([B]/[B]^\circ)^b}$ ↳ 1 mol/L

$K_p = \dfrac{(P_C/P_C^\circ)^c(P_D/P_D^\circ)^d}{(P_A/P_A^\circ)^a(P_B/P_B^\circ)^b}$ ↳ 1 bar = 10^5 Pa ☆

eg. $\dfrac{P_C : 2000\,Pa}{P_C^\circ : 10^5\,Pa} = 0.02$

4. Q & K can decide the direction of reaction

5. $pV = nRT \Rightarrow P/RT = n/V = c \Rightarrow$ c is directly proportional to p (at constant T)

6. the stronger the acid , the weaker its conjugate base

$HA \rightleftharpoons H^+ + A^-$

stronger　more　more

↳ weaker base (不易结合 H^+)

7. $pH = -\log(H^+)$

$K_w = [H^+][OH^-] = 10^{-14}$ (25°C)

$pKa = -\log(Ka)$

8. buffer range: pKa − 1 ~ pKa + 1

缓冲对中弱酸的解离常数的负对数

缓冲aq.的pH是否在其自身的缓冲范围内

9. 考的5类题型 pH calculation ⇒ 2a.p.

(1) 强酸&强碱

· 0.2 mol·dm⁻³ H_2SO_4

$[H^+] = 0.4\,mol/L \qquad pH = -\log[H^+] = -\log 0.4 = 0.40$

· 0.2 mol·dm⁻³ $Ca(OH)_2$

$[OH^-] = 0.4\,mol/L \qquad pH = 14 - pOH = 14 - (-\lg(0.4)) = 13.60$

⚠ : 几元酸/碱

(2) 弱酸的pH (Arrhenius definition)

· 0.2 mol·dm⁻³ CH_3COOH ($pKa = 4.76$)

↳ 先换成 $Ka = 10^{-4.76}$

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

0.2

x　　x　　x

0.2−x　　x　　x

$\dfrac{x^2}{0.2-x} = 10^{-4.76}$

$x = 1.86 \times 10^{-3} \therefore pH = -\lg(1.86 \times 10^{-3}) = 2.73$

· 0.2 mol·dm⁻³ H_3PO_4 ($pKa = 2.16, 7.20, 12.32$)

$Ka_1 = 10^{-2.16}$ magnitude stronger　negligible

$H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^-$

0.2

x　　x　　x

0.2−x　　x　　x

$\dfrac{x^2}{0.2-x} = 10^{-2.16}$

$x = 0.0339 \therefore pH = -\lg 0.0339 = 1.47$

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(3) Bronsted-Lowery definition

- $0.2\,mol\cdot dm^{-3}$ NH_4Cl (pK_a of $NH_4^+ = 9.25$)

$NH_4^+ \rightleftharpoons NH_3 + H^+$

0.2

x　　x　x

0.2−x　　x　x

$\frac{x^2}{0.2-x} = 10^{-9.25}$　$x = 1.06\times10^{-5}$　$pH = 4.97$

- $0.2\,mol\cdot dm^{-3}$ NH_3 (pK_a of $NH_4^+ = 9.25$)

$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

0.2

x　　x　x

0.2−x　　x　x

$K = \frac{[NH_4^+][OH^-]}{[NH_3]} = \frac{[NH_4^+]K_w}{[NH_3][H^+]} = 10^{9.25}\times10^{-14} = 10^{-4.75} = \frac{x^2}{0.2-x}$

$x = 1.87\times10^{-3}$　$\therefore 14-(-\lg 1.87\times10^{-3}) = 11.27$

(4) 共轭酸碱　混合后C改变

ethanoic acid (CH_3COOH, $0.2\,mol\cdot dm^{-3}$, $100\,cm^3$) and sodium ethanoate (CH_3COONa, $0.2\,mol\cdot dm^{-3}$, $100\,cm^3$) are mixed, giving solution A.

- calculate the "analytical concentration" of CH_3COOH (0.1 mol/L) and CH_3COONa (0.1 mol/L)

分析C：aq.中某物质的量除以总V，不考虑其在aq.中的解离

- calculate the pH of the solution A, using $pK_a = 4.76$

↳ $K_a = 10^{-4.76}$

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

0.1　　0.1

x　　x　x

0.1−x　　0.1+x　x

$\frac{x(0.1+x)}{0.1-x} = 10^{-4.76}$　$x = 1.74\times10^{-5}$　$pH = 4.76$

(5) 异性与同性aq.混合

- $5\,cm^3$ of $0.5\,mol\cdot dm^{-3}$ NaOH is added to solution A

A: HOAC: 0.02 mol, NaOAC: 0.02 mol → NaOH 2.5×10^{-3} mol, 0.0025

∴ NaOAC: $0.02 + 0.0025 = 0.0225\,mol = 2.25\times10^{-2}\,mol$

HOAC: $0.02 - 0.0025 = 0.0175\,mol = 1.75\times10^{-2}\,mol$

↳ 用n算反应后的各个物质的量（看明面上的物质写I）

$HOAC \rightleftharpoons H^+ + AcO^-$

I: $\frac{1.75\times10^{-2}}{205\times10^{-3}}$　0　$\frac{2.25\times10^{-2}}{205\times10^{-3}}$

C: −x　　x　x

E: ..　　..　..

$x = 1.3516\times10^{-5}$

$pH = 4.87$

↳ 转换成C △V的变化与单位

- 200 ml H_2O : pH = 7

↑ 0.5 mol/L的5 ml NaOH

$0.5\times5\times10^{-3} = 2.5\times10^{-3}\,mol$

$pOH = -\lg\left(\frac{2.5\times10^{-3}}{0.205}\right) = 1.913$　$pH = 12.08$

- $5\,cm^3$ of $0.5\,mol\cdot dm^{-3}$ HCl is added to solution A

A: HOAC: 0.02 mol, NaOAC: 0.02 mol → HCl 2.5×10^{-3} mol, 0.0025 mol　强酸与和弱酸盐反应（弱酸根 ≈ base）

∴ NaOAC = 1.75×10^{-2} mol　HOAC = 2.25×10^{-2} mol

$HOAC \rightleftharpoons H^+ + AcO^-$

I: $\frac{2.25\times10^{-2}}{205\times10^{-3}}$　0　$\frac{1.75\times10^{-2}}{205\times10^{-3}}$

C: −x　　x　x

E: ...

$\frac{\left(\frac{1.75\times10^{-2}}{205\times10^{-3}}+x\right)x}{\left(\frac{2.25\times10^{-2}}{205\times10^{-3}}-x\right)} = 10^{-4.76} = \frac{(0.085+x)x}{0.10975-x} = 10^{-4.76}$

$0.085x + x^2 = 1.9072\times10^{-6} - 10^{-4.76}x$　$x_1 = 2.24\times10^{-5}$　$x_2 < 0$

$pH = -\lg(2.24\times10^{-5}) = 4.65$

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Chap 7 Thermochemistry and Thermodynamics 考点

1. adiabatic - 绝热的 piston 活塞 diabatic 非绝热的
   gas cylinder 气瓶 stopper 塞子 non-insulated 不隔热的

2. $\Delta U = q + w$ [unit of each is joule, J] , U can't be measured
   always f-i
   q, w → path function
   for s.l. and ideal gas. U is only a function of T and n → U is constant for isothermic process → $\Delta U_{isothermal} = 0$
   等T过程

   $q = cm\Delta T$ (物态不发生变化) c is the specific heat capacity of a subtance
   记: $C(H_2O) = 4.18\ J \cdot g^{-1} \cdot K^{-1}$

3. work with constant P: $w = -p_{ext}\Delta V = -p_{ext}(V_f - V_i)$
   Changing : $w = -\int_{V_i}^{V_f} p_{ext}\, dV = -nRT \ln V_f/V_i$
   sign ☆

$H = U + pV$

4. isobaric process:
   $\Delta U = q + w = q_p - p\Delta V$
   - q_p is the heat of reaction at constant pressure, can be experimentally measured.
   - $q_v = \Delta U$

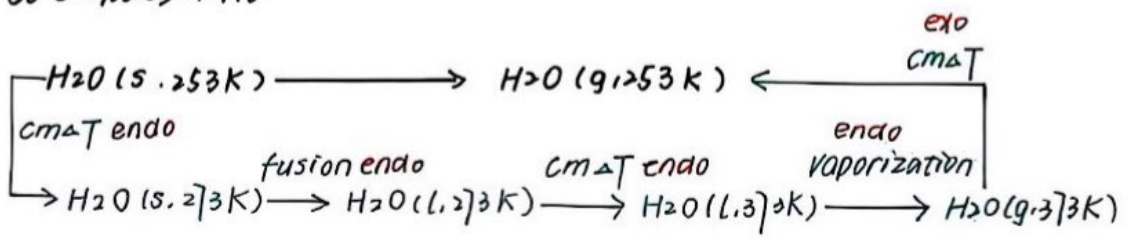
   $q_p = \Delta U + p\Delta V = (U_f - U_i) + p(V_f - V_i) = H_f - H_i = \Delta H$ (equal numerically in pec)

5. ΔH° ↘ standard enthalpy change unit: kJ/mol
   - 有气体物质分压为 1 bar
   - for solution: C = 1mol/L
   - allotrops: graphite, O_2 (most stable)
     exp. white phosphorus (less stable than red)

6. Enthalpy change for $\boxed{\text{Physical Transformations}}$

$H_2O(l) \rightarrow H_2O(g)$	$\Delta H^\circ_{vap} = +40.66\ kJ \cdot mol^{-1}$	at 373K
$H_2O(g) \rightarrow H_2O(l)$	$\Delta H^\circ_{con} = -40.66\ kJ \cdot mol^{-1}$	at 373K
$H_2O(s) \rightarrow H_2O(l)$	$\Delta H^\circ_{fus} = +6.01\ kJ \cdot mol^{-1}$	at 273K
$H_2O(l) \rightarrow H_2O(s)$	$\Delta H^\circ_{freez} = -6.01\ kJ \cdot mol^{-1}$	at 273K
$I_2(s) \rightarrow I_2(g)$	$\Delta H^\circ_{sub} = +62.42\ kJ \cdot mol^{-1}$	at 385K.

考: calculate ΔH°_{sub} of ice at 253K
average heat capacity ($J \cdot g^{-1} \cdot K^{-1}$): ice (-50°C ~ 0°C): 2.108; water (0°C ~ 100°C): 4.187; water vapour (-50°C ~ 100°C): 1.996



H is a state function
the 2 path have the same initial & final state

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7. Enthalpy of Atomic and Molecular Change

Standard Enthalpy of Atomization, ΔH°_{atom}

when 1 mole of the gaseous atoms are formed from its elements under standard conditions

$Na(s) \rightarrow Na(g)$ $\quad$ $1/2\,H_2(g) \rightarrow H(g)$

$1/2\,I_2(s) \rightarrow I(g)$

Standard Enthalpy of Ionization, ΔH°_{ion}

the romoval of an electron from a gas-phase atom or ion [Subsequent electron may be removed after the first]

$Mg(g) \rightarrow Mg^+(g)$ $\quad$ $\Delta H^\circ_{ion.1st} = 738\ kJ\cdot mol^{-1}$

$Mg^+(g) \rightarrow Mg^{2+}(g)$ $\quad$ $\Delta H^\circ_{ion.2nd} = 1451\ kJ\cdot mol^{-1}$

Electron Affinity, ΔH°_{EA}

when 1 mole of gaseous atoms/ions accepts 1 mole of electrons to form 1 mole of anions

考~符号

- for most $X + e^- = X^-$ is exo (except noble gases, nitrogen (endo) & a few metals)
- for all X^{2-}.... is endo

$O(g) + e^- \rightarrow O^-(g)$ $\quad$ $\Delta H^\circ_{EA} = -142\ kJ\cdot mol^{-1}$

$O^-(g) + e^- \rightarrow O^{2-}(g)$ $\quad$ $\Delta H^\circ_{EA} = +844\ kJ\cdot mol^{-1}$

Lattice Energy, ΔH°_{lat}

when 1 mole of the crystalline substance is formed from its gaseous ions

$Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$ $\quad$ $\Delta H^\circ_{lat} = -774\ kJ\cdot mol^{-1}$

8. Enthalpy of Chemical Change

Standard Enthalpy of Combustion $\Delta_c H^\circ$

when 1 mole of the pure substance is completely burnt

$\Delta H = q_v + (\Delta n)RT$ $\quad$ Δn is the difference of gas molecules in the balanced equation

Standard Enthalpy of Formation, $\Delta_f H^\circ$

when 1 mole of the pure substance is formed from its elements under standard conditions

只有元素最稳定的单质形态的生成焓为0 $\quad$ H_2、N_2、O_2、Hg、Na、graphite、Fe

9.

BDE

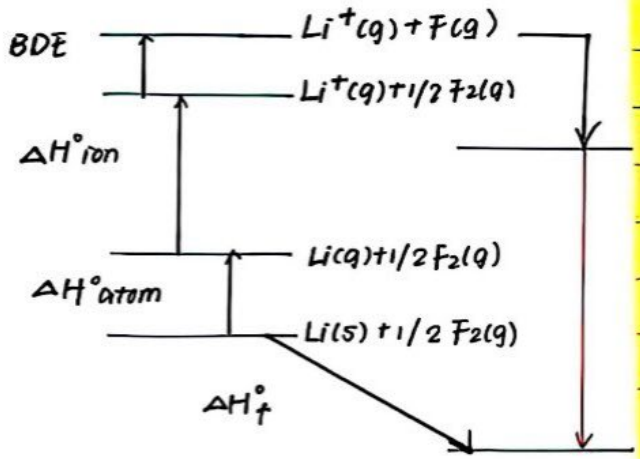
breaking a bond to form atoms

more than 1 bond, then take the avera

10 $\Delta_r H^\circ = \sum \Delta_f H^\circ_{pro} - \sum \Delta_f H^\circ_{reac}$ → 一定不一初

$\Delta_r H^\circ = \sum BDE(bro) - \sum BDE(form)$

考 Born-Haber cycle



Born-Haber 循环总结

步骤	名称	公式	相变
1	升华焓	$M(s) \rightarrow M(g)$	固→气
2	电离能	$M(g) \rightarrow M^+(g) + e^-$	X
3.	汽化/升华焓(键)	$X_2(l/s) \rightarrow X_2(g)$	液/固→气
4.	BDE	$1/2\,X_2(g) \rightarrow X(g)$	分子断裂
5.	电子亲和能	$X(g) + e^- \rightarrow X^-(g)$	X
6.	晶格能	$M^+(g) + X^-(g) \rightarrow MX(s)$	气→固
7.	生成焓	元素→$MX(s)$	总反应

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when 1 mole of the gaseous atoms are formed from its elements under standard condi

$Na(s) \rightarrow Na(g)$ $\quad$ $1/2\,H_2(g) \rightarrow H(g)$

$1/2\,I_2(s) \rightarrow I(g)$

Standard Enthalpy of Ionization, ΔH°_{ion}

the removal of an electron from a gas-phase atom or ion [Subsequent electron n be removed after the first]

$Na^+(g)$ $\quad \Delta H^\circ_{ion.1st} = 738\ kJ\cdot mol^{-1}$

$\Delta H^\circ_{ion.2nd} = 1451\ kJ\cdot mol^{-1}$

ns accepts 1 mole of electrons to form 1 mole of anio

exo (exoth noble gases nitrogen & a few metal endo

is endo

$\Delta H^\circ_{EA} = -14$

ΔH

mical Change

alpy of Combustion $\Delta_c H$

1 mole of the pure substance

$H = q_p + (\Delta n$ $\quad$ Δn is the ch

tion. $\Delta_f H^\circ$

pure substance

天然形态的生成焓为

king a bond to form atoms

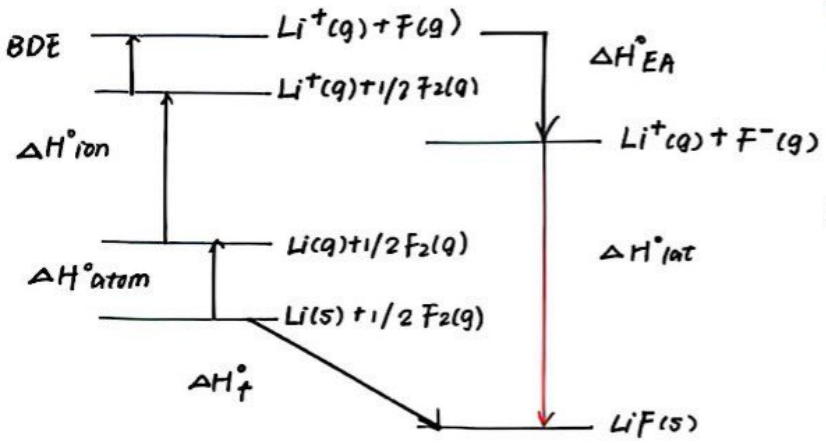
more than 1 bond, then take the average value.

$\Delta_r H^\circ = \sum \Delta_f H^\circ_{pro} - \sum \Delta_f H^\circ_{reac}$ 一定来初

multiply individual $\Delta_f H^\circ$ by stoichiometric num

$\Delta_r H^\circ = \sum BDE(bro) - \sum BDE(form)$

Born-Haber cycle



物质状态的确定:

元素	在298K下的 standard state
Na, K, Mg, Ca	s
Cl_2, F_2, O_2, N_2, H_2	g
Br_2	l
I_2	s
S_8, P_4	s

必须先考虑汽化或升华步骤

拆保到单原子.

Chap 8 2nd Law. Entropy & ΔG

1. micro: $S = k_B \ln \Omega$ → $\frac{R}{N_A}$

2. macro:

Comparison

ΔS表达式	适用情景	ΔS含义	单位
$\int \frac{C_p}{T} dT$	Δ/冷却单一物质	每1mol物质的熵变	$J \cdot mol^{-1} \cdot K^{-1}$
$\Sigma S^\circ_{pro} - \Sigma S^\circ_{rea}$	化学反应	每1mol反应方程式的熵变	$J \cdot mol^{-1} \cdot K^{-1}$

单一物质:

reversible: $\Delta S = \frac{\Delta q_{rev}}{T}$ or $dS = \frac{\delta q_{rev}}{T}$ 只在可逆过程中成立,若过程不可逆,只能用积分的"可逆路径"来计算ΔS

irreversible: per mole $\Delta S = \int_{T_i}^{T_f} \frac{C(T)}{T} dT$ (per mole) unit: $\frac{J \cdot mol^{-1} \cdot K^{-1}}{K} \cdot K = J \cdot mol^{-1} \cdot K^{-1}$

if C is a constant, $\Delta S = C \ln(T_2/T_1)$ from T_1 to T_2

3. standard entropy: S°_m [在1bar压力下,1mol(通常最稳定形态的)物质在某温度(通常是298.15K)的绝对熵值] $T_i = 0K$, $S_i = 0$

化学反应: $\Delta_r S^\circ = \Sigma S^\circ_{pro} - \Sigma S^\circ_{rea}$ ☆ remember the stoichiometric coefficients

☆ $\Delta_f H^\circ (kJ \cdot mol^{-1})$ $S^\circ_m (J \cdot K^{-1} \cdot mol^{-1})$ → entropy per mole of substance (↳ mole)

4. entropy change for phase transition

$\Delta S_{fus} = \frac{\Delta H_{fus}}{T_m}$ $\Delta S_{vap} = \frac{\Delta H_{vap}}{T_b}$

5. Gibbs Free Energy

entropy change in the surrounding =

(1) $\Delta S_{sur} = \frac{q_{sur,rev}}{T}$ → the heat into the surroundings

$\Delta q_{sur,rev} = \Delta H_{sur} = \Delta q_{sur,irre}$ (state function)

$\therefore \Delta S_{sur} = \frac{q_{sur}}{T}$

(2) $q_{sur} = -q_{sys} = -q$ 可逆 $\therefore \Delta S_{sur} = -\frac{q_{sys}}{T_{sys}}$ (re & irre)

(3) isobaric $q = q_p = \Delta H$ $\therefore \Delta S_{sur} = -\frac{\Delta H_{sys}}{T_{sys}}$

$\Delta S_{sur} = -\frac{\Delta H}{T}$ therefore,

$\Delta S_{uni} = \Delta S_{sys} + \Delta S_{sur} = \Delta S_{sys} - \frac{\Delta H_{sys}}{T}$

$T\Delta S_{uni} = T\Delta S_{sys} - \Delta H_{sys} = -\Delta G_{sys}$ $\Delta G = \Delta H - T\Delta S$ ☆

Application

① $\Delta G \begin{cases} < 0 & \text{spo} \\ = & e \\ > & \text{non-spon} \end{cases}$ ② $\Delta G = W'_{max}$ (X p-V work)

$\Delta G = \Delta H - T\Delta S$ → unit: K 不同的T对应了不同的ΔG

考 independent of T. 可看作 constant. 可用298K来 calculate

6. How to calculate $\Delta_r G^\circ$

① $\Delta_r G^\circ = \Sigma \Delta_f G^\circ_{products} - \Sigma \Delta_f G^\circ_{reactants}$ 是不是要求的T 元素最稳定的单质形态 $\Delta_f H^\circ = 0$, $\Delta_f G^\circ = 0$

② $\Delta_r G^\circ = \Delta_r H^\circ - T\Delta_r S^\circ$

$kJ \cdot mol^{-1}$ $kJ \cdot mol^{-1}$ $J \cdot mol^{-1}$ 单位时刻带好,记得换算单位

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考7. Δ_rG°. Δ_rG & K

$\Delta_rG = \Delta_rG^\circ + RT\ln Q$ (K)

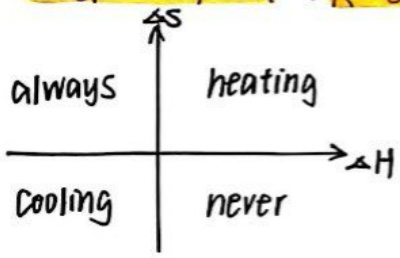
$kJ\cdot mol^{-1}$ $kJ\cdot mol^{-1}$ $J\cdot mol^{-1}\cdot K^{-1}$ → 换算单位 $\times 10^{-3}$

$Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$ ① IUPAC ② 若是气体，是否已是分压

$0 = \Delta_rG = \Delta_rG^\circ + RT\ln K$ [when equilibrium is established]

$\Delta_rG^\circ = -RT\ln K$. $K = e^{-\frac{\Delta_rG^\circ}{RT}}$ 统一单位

8.



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Chap 9 Organic Chemistry: Hydrocarbon and Isomerism

1. IUPAC Nomenclature rule

① Maximum number of principle functional group cited as suffix

② Maximum number of C=C and C≡C bonds together

③ Maximum length of parent chain/ring

④ Maximum number of C=C bonds

⑤ lowest locants for principle groups

⑥ lowest locants for multiple bonds

⑦ lowest locants for double bonds

⑧ Maximum number of substituents cited as prefix

⑨ Lowest locants for the substituents cited as prefix

⑩ Lowest locant for the first cited subtituents

2. Alkane C_nH_{2n+2}

g: $C_{1\sim4}$: methane ethane propane butane

l: $C_{5\sim14}$: pentane hexane heptane octane nonane decane undecane dodecane

⋮

Prefix ~ (num. of identical sub.): di- tri- tetra- ···

Substituents are named by changing '-ane' to '-yl'.

methyl	$-CH_3$	Me
ethyl	$-CH_2CH_3$	Et
propyl	$-CH_2CH_2CH_3$	Pr
butyl	$-(CH_2)_3CH_3$	Bu

3. 作图常用skeletal structure，只有与C相连的H不用画

千万别忘了

4. Cycloalkanes are named by adding the prefix cyclo- to the name of the alkane C_nH_{2n}

5. Substituents are in alphabetical order，最后一个取代基与主链无空格

halo groups (fluoro, chloro, bromo, iodo)

从停在parent chain的那一端开始编号

可加数词前缀

6-(3-bromopropyl)-2,3-dimethyldecane

空格

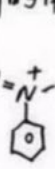
parentheses：代表这是一个在位置6上的取代基，

6. 多取代基编号⇒逐位比较大小

编号相同→字母前的用编号小的

7. Alkene C_nH_{2n}

DBE = Ω



$-NO_2$ ⇒ Ω=1

>C=C< Ω=1

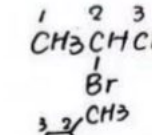
-C≡C- Ω=2

⬡ Ω=1

⬡ Ω=4

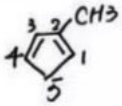
$$DBE = \frac{C\times2+2-H-X+N}{2}$$ (0.5不内入)

compounds with 2, 3, ·· C=C bonds are named as "diene", "triene"



2-bromo-4-methylhex-3-ene

捕不对捕音



2-methylcyclopenta-1,3-diene

8.

isopropyl benzene 异丙苯

n-butane 正丁烷

isobutane 异丁烷

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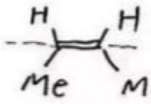
9. cis-trans isomer



沿double bond切，相同原子在一侧 ~ cis
异 ~ trans

trans-pent-2-ene

cis-but-2-ene (顺-2-丁烯)

cis and trans isomers are not possible if 2 substituents on an sp^2 carbon are the same

10. → H,H异侧

Z-E Nomenclature System

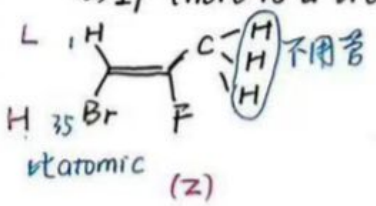
H,H同侧

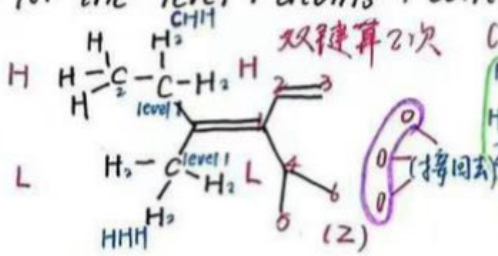
Cahn-Ingold-Prelog (CIP) Rule

(1) Compare the atomic number of the two atoms which are directly attached to the C=C carbon. For ease of discussion, there're the "level 1" atoms. The atom with higher atomic number has higher priority. eg. Cl > O > N > C > H

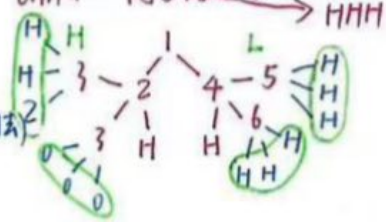
(2) If level 1 atoms are isotopes of the same element, compare their mass number. The isotope with higher mass number has higher priority. eg. D > H

(3) If there is a tie for the "level 1 atoms", continue ·· 2 ··, ·· 3 ·· if needed.





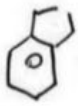
双键算2次



(每层比大小都是原子序从大→小依次比)

11. Nomenclature of other organic compounds

				eg.
alcohol	R-OH	-OH (hydroxyl)	-ol	propanol
aldehyde	R-CHO	-CO- (carbonyl)	-al	propanal
ketone	R^1-CO-R^2	-CO- (carbonyl)	-one	propanone
carboxylic acid	R-COOH	-COOH (carboxyl)	-oic acid	propanoic acid
arene	R-C_6H_5	benzene	benzene	diethylbenzene
phenol	R-C_6H_4-OH	phenol	phenol	2-chlorophenol



1,2-
ortho

1,3-
meta

1,4-
para

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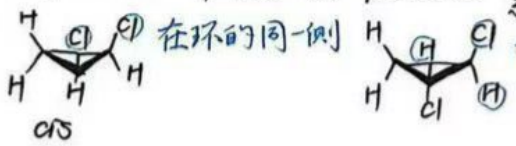
Chap 10 Chirality 手性

1. isomerism
   - constitutional isomers
     - chain isomerism / skeleton isomerism
     - positional isomerism
     - functional group isomerism
   - stereoisomers
     - enantiomers (mirror-image)
     - diastereomers (non-mirror-image)
       - configurational diastereomers
       - cis-trans diastereomers

2. Isomers of pentane 石戊烷: n-pentane 异戊烷: iso-pentane 新戊烷: tert-pentane

环也有 cis-trans diastereomers



3. chirality:
   (is non-superimposable) with its mirror image
   enantiomer:
   (a pair of mirror images that are non-superimposable)
   diastereomer:
   (non-superimposable & non-mirror-images)

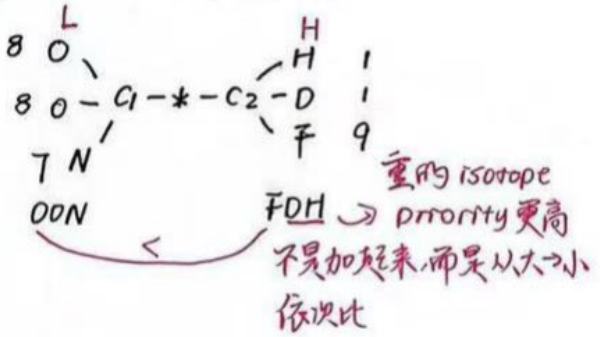
4. chiral center is a tetrahedral atom bonded to 4 different groups.
   ↳ can be various elements, but most common is an sp^3 carbon, marked with an asterick *

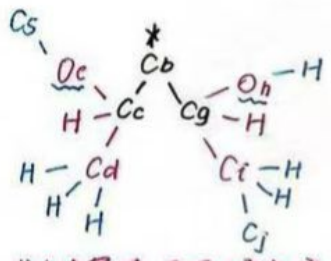
5. ▬ point towards the observer
   ┆┆┆ away from
   By convention, the ring or longest carbon chain is on solid lines
   multiple ways to draw the same enantiomer: 一笔三转

考 ⑥ Absolute configuration 绝对构型
☆ (只与 molecular structure 有关)

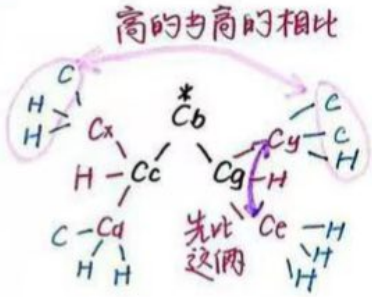
- Step 0: Identify the chiral center and find out the 4 different groups attach to the *
- Step 1: Rank the 4 groups by CIP rule: the group with the highest priority is ranked lowest
- Step 2: Orient the molecule such that the lowest priority group (4) is pointing away from the observer
- Step 3: Count the direction of priority group 1→2→3 { 顺时针 R / 逆时针 S

☆ CIP 重点说明 (易错)

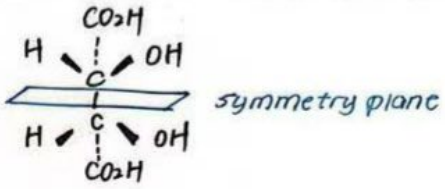




谷文先算前一层原子序数高的所连接的原子比



8. Meso compound (内消旋化合物)



It's an "achiral" molecule but there're chiral centers within the molecule

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